CurenZee

A Mobile Based System for Real Time Sri Lankan Currency Recognition using Object Detection: An Accessible Solution for the Visually Impaired

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AGENDA

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Introduction

Problem Associated

- Visually impaired individuals face many difficulties in recognising currency denominations.
- Manual identification methods such as tactile features are inadequate.
- Risk of easily being misled, fraud, or overcharged (Mr. Chandrashekhar Mankar et al., 2023).
- Only a limited number of existing solutions exist, and they have limitations (Perera and Meedin, 2023).

Research Gap

- Most of the existing solutions are limited to note identification, and solutions that address coins are very low in accuracy (Abimani et al., 2021).
- Existing solutions are only limited to daylight identification and do not support low-light identification (Gamage et al., 2020).
- Many existing solutions do not provide real-time recognition with multiple denomination detection (R.N, 2022).
- Most solutions do not provide a user-friendly application with useful accessibility features to overcome the issues of visually impaired individuals (Gamage et al., 2020).

Project's Aim

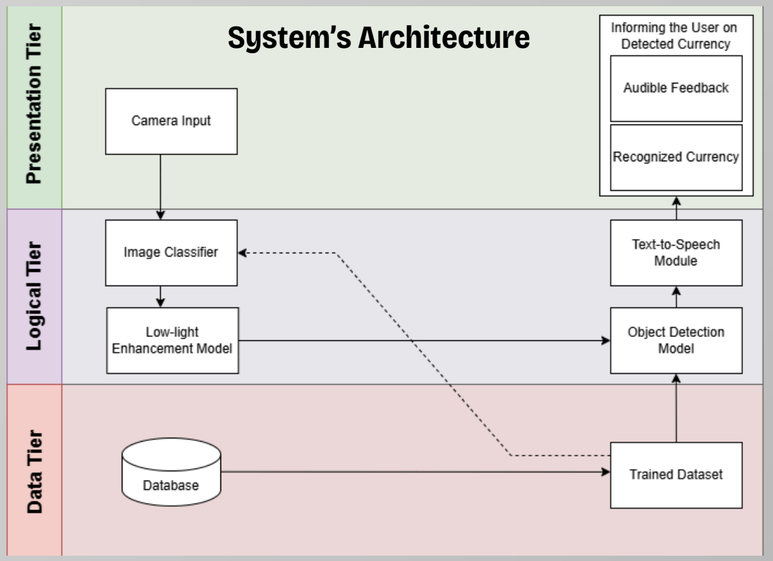
Research Aim

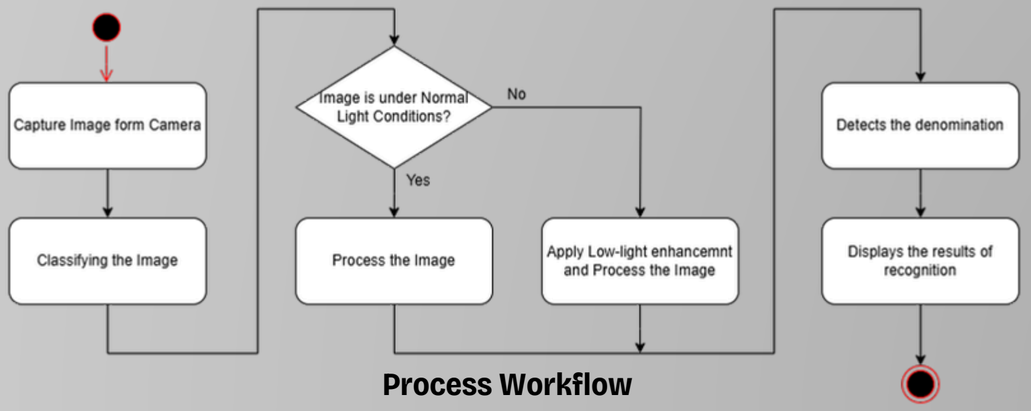
- Developing a mobile application to accurately detect and classify Sri Lankan currency denominations in real-time using deep learning.

Objectives

- Implementing a system with YOLOv8 and CLAHE for currency recognition.
- Multiple denomination detection in varied light conditions in real time.
- Developing an acessible Flutter-based mobile application with audible feedback feature.

Design of the system





Process Workflow

Implementation

Technology Stack

Core Functionalities

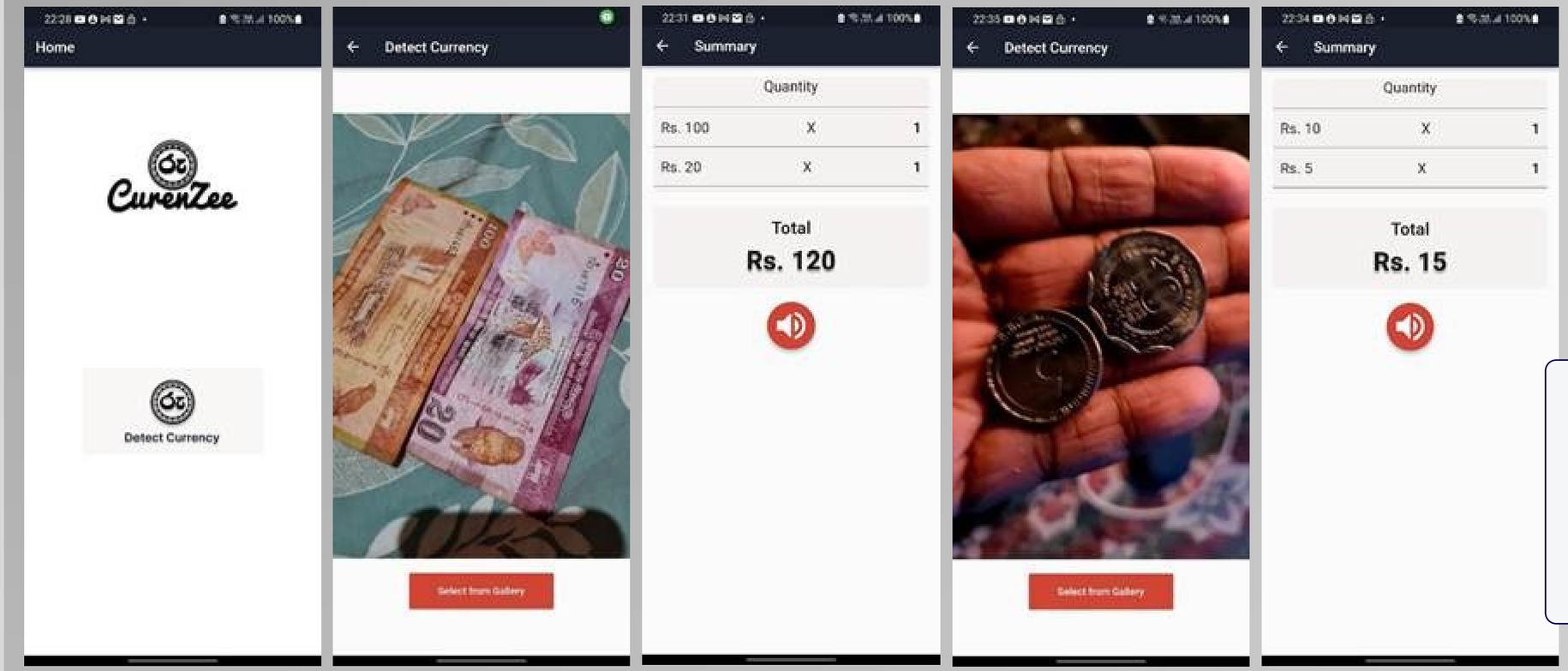
Backend Processing

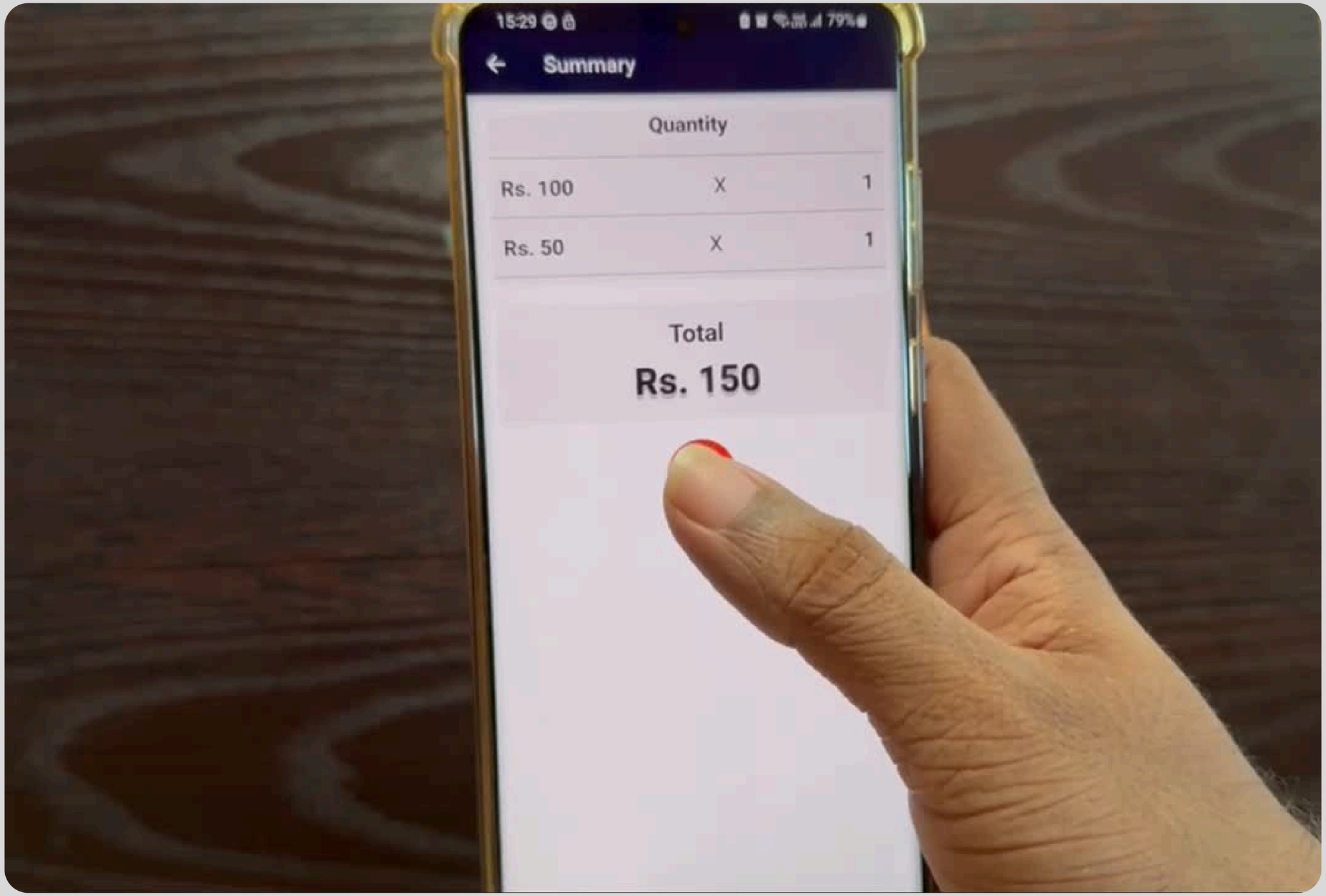
- **Currency Recognition:** YOLOv8 detects and classifies Sri Lankan currency notes & coins.
- **Low-Light Enhancement:** CLAHE enhances image clarity for better recognition.
- **Text-to-Speech (TTS):** Converts detected currency into audio feedback.

Frontend Features

- **Image Capture:** The user takes a photo via the mobile app.
- **Real-Time Recognition:** The model processes and displays results.
- **Audio Feedback:** Identified currency is read aloud for accessibility.

Demonstration

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79%
Summary
Quantity
Rs. 100 X 1
Rs. 50 X 1
Total
Rs. 150

Testing and Evaluation

Functional Testing Results

- Unit Testing: Tested the individual modules.
- Intergration Testing: Ensured seamless communication between frontend and backend.
- System Testing: Conducted end-to-end testing with real-world currency images.

Non-Functional Testing Results

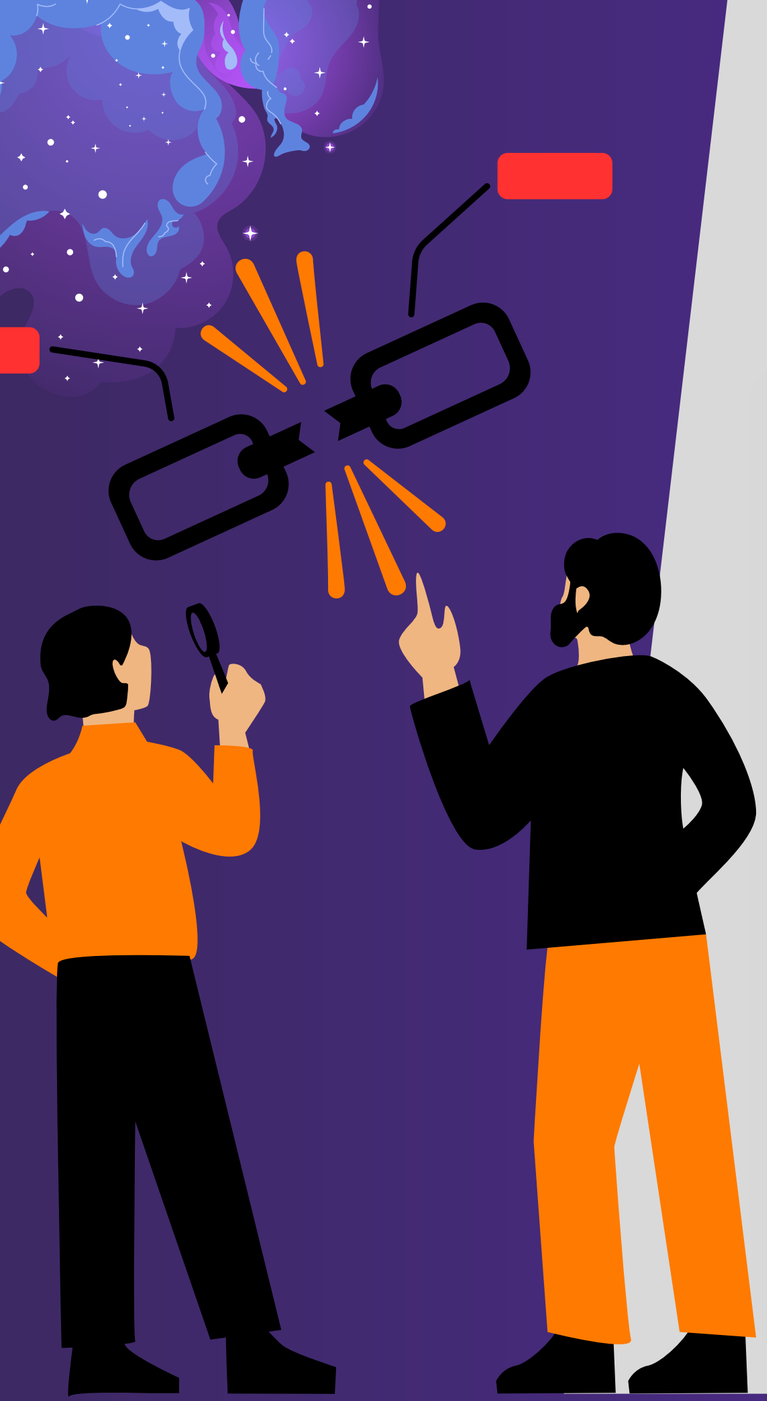
- Response Time: Less than 1s Identification.
- Reliability: System functions correctly under varying lighting conditions

Challenges Faced

- Handling Low-Light & Blurry Images: Implemented CLAHE for image enhancement & improved preprocessing
- Ensuring Real-Time Performance: Using YOLOv8 for identification with optimizations.
- Achieving high accuracy: fine-tuning the hyperparameters and optimizing the model.

Significance & Novelty

- Empowers vision-impaired individuals with independent currency recognition.
- Utilizes YOLOv8 and CLAHE to implement a unique, effective solution.
- Identifying both Sri Lankan currency notes and coins under varied environmental conditions.
- Enables real-time multiple currency denomination detection.



Limitations and Future work

Limitations

- Unable to recognise very blurred, damaged, or worn denominations under extreme conditions.
- Limited dataset with varied environmental light conditions.

Future work

- Expansion of the dataset with various extreme light conditions.
- Integration of more advanced deep learning techniques to handle the extreme conditions, including low light and wear and tear & using YOLOv11 for image recognition.
- Integration of voice commanding facility.
- Feature to view previous records.

Conclusion

Achivements

- Accurate, real-time currency note and coin recognition in varying lighting conditions.
- Flutter-based mobile application with a user-friendly interface for real-time recognition.

Skills accuired

New Skills Acquired	Existing Skills Used
• Model training and evaluation • API integration with Flask • Hands-on experience with object detection and image enhancement • Identifying a proper research gap and implementing a practical solution to it	• Python programming • Mobile app development • Project management skills • Research methodology and academic writing.

Summary

Problem Addressed

- Challenges and difficulties vision impaired individuals face in Sri Lankan currency identification

Proposed Solution

- Implement a mobile application that enables vision-impaired individuals to accurately identify notes and coin denominations of Sri Lanka in real-time.

Key Features

- Using YOLOv8, logistic regression, and CLAHE to implement a unique, effective solution.
- Both notes and coins are identified under varied environmental conditions.
- Performs in real-time with multiple denomination detection.
- User-friendly interface with accessibility features like audible feedback

Thank You